

ECA1803 - OPTOELECTRONICS & OPTICAL COMMUNICATION

Total points 48/50 ?

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✓ The electron hole pairs generated in a photodiode are separated by the *1/1

- ☐ a) Magnetic field
- ☒ b) Electric field
- ☐ c) Static field
- ☐ d) Depletion region



SLOT *

☒ c



✓ Light incident on fibers of angles _____ the acceptance angle do not propagate into the fiber *1/1

- ☐ a) Less than
- ☒ b) Greater than
- ☐ c) Equal to
- ☐ d) Less than and equal to



✓ As compared to planar LED structure, Dome LEDs have _____ External power efficiency _____ effective emission area and _____ radiance *1/1

- ☐ a) Greater, lesser, reduced
- ☒ b) Higher, greater, reduced
- ☐ c) Higher, lesser, increased
- ☐ d) Greater, greater, increased



✓ Which of the following detector is fabricated from semiconductor alloys? * 1/1

- ☒ a) Photoconductive detector
- ☐ b) p-i-n detector
- ☐ c) Photodiodes
- ☐ d) Photoemission detectors



✓ In modified chemical vapor deposition, vapor phase reactant such as _____ pass through a hot zone. *1/1

- ☒ a) Halide and oxygen
- ☐ b) Halide and hydrogen
- ☐ c) Halide and silica
- ☐ d) Hydroxides and oxygen



✓ An optical fiber has a core radius $2\mu\text{m}$ and a numerical aperture of 0.1. Will this fiber operate at single mode at 600 nm? *1/1

- ☒ a) Yes
- ☐ b) No



✓ _____ technique is method of preparing extremely pure optical glasses. *1/1

- ☐ a) Liquid phase (melting)
- ☐ b) Radio frequency induction
- ☐ c) Optical attenuation
- ☒ d) Vapor Phase Deposition (VPD)



✓ The depletion region must be _____ to allow a large fraction of the incident light to be absorbed in the device(photodiode). *1/1

- ☒ a) Thick
- ☐ b) Thin
- ☐ c) Long
- ☐ d) Inactive



✓ The majority of the carriers in a p-type semiconductor are _____ * 1/1

- ☒ a) Holes
- ☐ b) Electrons
- ☐ c) Photons
- ☐ d) Neutrons



✓ The quantum efficiency of an optical detector should be high. * 1/1

- ☒ a) True
- ☐ b) False



✓ The fraction of incident photons generated by photodiode of electrons generated collected at detector is known as _____ *1/1

- ☒ a) Quantum efficiency
- ☐ b) Absorption coefficient
- ☐ c) Responsivity
- ☐ d) Anger recombination



✓ Multimode step index fiber has a large core diameter of range is * 1/1

- ☒ a) 100 to 300 μm
- ☐ b) 100 to 300 nm
- ☐ c) 200 to 500 μm
- ☐ d) 200 to 500 nm



✓ Single mode fibers allow single mode propagation; the cladding diameter must be at least *1/1

- ☐ a) Twice the core diameter
- ☐ b) Thrice the core diameter
- ☐ c) Five times the core diameter
- ☒ d) Ten times the core diameter



✓ Multimode step index fibers have a bandwidth of *

1/1

- ☐ a) 2 to 30 MHz km
- ☒ b) 6 to 50 MHz km
- ☐ c) 10 to 40 MHz km
- ☐ d) 8 to 40 MHz km



SUBJECT CODE *

- ☒ ECA1803

✓ The width of depletion region is dependent on _____ of semiconductor.

*1/1

- ☒ a) Doping concentrations for applied reverse bias
- ☐ b) Doping concentrations for applied forward bias
- ☐ c) Properties of material
- ☐ d) Amount of current provided



✓ The ratio of speed of light in air to the speed of light in another medium is *1/1 called as

- ☐ a) Speed factor
- ☐ b) Dielectric constant
- ☐ c) Reflection index
- ☒ d) Refraction index



✗ Which of the following does not explain the requirements of an optical detector? *0/1

- ☐ a) High quantum efficiency
- ☒ b) Low bias voltages
- ☐ c) Small size
- ☐ d) Low fidelity



Correct answer

- ☒ d) Low fidelity

✓ The photocurrent of an optical detector should be _____ * 1/1

- ☐ a) Less
- ☐ b) More
- ☒ c) Linear
- ☐ d) Non-linear



✓ Silicon has indirect band gap energy of _____ * 1/1

- ☐ a) 1.2 eV
- ☐ b) 2 eV
- ☒ c) 1.14 eV
- ☐ d) 1.9 eV



✓ The difference between the modes' refractive indices is called as * 1/1

- ☐ a) Polarization
- ☐ b) Cutoff
- ☒ c) Fiber birefringence
- ☐ d) Fiber splicing



✓ A measure of amount of optical fiber emitted from source that can be coupled into a fiber is termed as _____ *1/1

- ☐ a) Radiance
- ☐ b) Angular power distribution
- ☒ c) Coupling efficiency
- ☐ d) Power-launching



✓ Multimode step index fiber has *

1/1

- ☒ a) Large core diameter & large numerical aperture
- ☐ b) Large core diameter and small numerical aperture
- ☐ c) Small core diameter and large numerical aperture
- ☐ d) Small core diameter & small numerical aperture



✓ The carrier recombination lifetime becomes majority or injected carrier lifetime.

*1/1

- ☐ True
- ☒ False



✓ Multimode graded index fibers with wavelength of $0.85\mu\text{m}$ have numerical aperture of 0.29 have core/cladding diameter of

*1/1

- ☐ a) $62.5\mu\text{m}/125\mu\text{m}$
- ☒ b) $100\mu\text{m}/140\mu\text{m}$
- ☐ c) $85\mu\text{m}/125\mu\text{m}$
- ☐ d) $50\mu\text{m}/125\mu\text{m}$



✓ The ratio $r = (n_1 - n)/(n_1 + n)$ indicates *

1/1

- ☐ a) Fresnel reflection
- ☒ b) Reflection coefficient ✓
- ☐ c) Refraction coefficient
- ☐ d) Angular power distribution coefficient

✓ Determine the total carrier recombination lifetime of a double heterojunction LED where the radioactive and nonradioactive recombination lifetime of minority carriers in active region are 70 ns and 100 ns respectively.

*1/1

- ☒ a) 41.17 ns ✓
- ☐ b) 35 ns
- ☐ c) 40 ns
- ☐ d) 37.5 ns

✓ The insertion losses of the fiber splices are much less than the Fresnel reflection loss at a butted fiber joint.

*1/1

- ☒ a) True ✓
- ☐ b) False



✓ Multimode graded index fibers are manufactured from materials with * 1/1

- ☐ a) Lower purity
- ☒ b) Higher purity than multimode step index fibers.
- ☐ c) No impurity
- ☐ d) Impurity as same as multimode step index fibers.



✓ A permanent joint formed between two different optical fibers in the field *1/1 is known as a

- ☒ a) Fiber splice
- ☐ b) Fiber connector
- ☐ c) Fiber attenuator
- ☐ d) Fiber dispersio



✓ _____ is used when the optical emission results from the application of electric field. *1/1

- ☐ a) Radiation
- ☐ b) Efficiency
- ☒ c) Electro-luminescence
- ☐ d) Magnetron oscillator



✓ The performance characteristics of multimode graded index fibers are * 1/1

- ☒ a) Better than multimode step index fibers
- ☐ b) Same as multimode step index fibers
- ☐ c) Lesser than multimode step index fibers
- ☐ d) Negligible



✓ Which of the following materials is not used as a starting material in vapor-phase deposition technique? *1/1

- ☐ a) SiCl_4
- ☐ b) GeCl_4
- ☐ c) O_2
- ☒ d) B_2O_3



✓ Multimode graded index fibers use incoherent source only * 1/1

- ☐ a) True
- ☒ b) False



NAME OF THE STUDENT *

MUDE NILESH NAIK



✓ The process of excitation of an electron from valence band to conduction band leaves an empty hole in the valence band and is called as _____

- ☐ a) Detection
- ☐ b) Absorption
- ☐ c) Degeneration of an electron-hole pair
- ☒ d) Regeneration of an electron-hole pair



✓ What is the numerical aperture of the fiber if the angle of acceptance is 16 degree? *1/1

- ☐ a) 0.50
- ☐ b) 0.36
- ☐ c) 0.20
- ☒ d) 0.27



✓ The first generation systems of optical fiber communication have wavelengths between _____ *1/1

- ☐ a) 0.2 and 0.3 μm
- ☐ b) 0.4 and 0.6 μm
- ☒ c) 0.8 and 0.9 μm
- ☐ d) 0.1 and 0.2 μm



✓ The absence of _____ in LEDs limits the internal quantum efficiency. *1/1

- ☐ a) Proper semiconductor
- ☐ b) Adequate power supply
- ☒ c) Optical amplification through stimulated emission ✓
- ☐ d) Optical amplification through spontaneous emission

✓ Standard single mode fibers (SSMF) are utilized mainly for operation in * 1/1

- ☐ a) C-band
- ☐ b) L-band
- ☒ c) O-band ✓
- ☐ d) C-band and L-band

✓ When a ray of light enters one medium from another medium, which quality will not change *1/1

- ☐ a) Direction
- ☒ b) Frequency ✓
- ☐ c) Speed
- ☐ d) Wavelength



✓ The light sources used in fibre optics communication are *

1/1

- ☒ a) LED's and Lasers
- ☐ b) Phototransistors
- ☐ c) Xenon lights
- ☐ d) Incandescent



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✓ Optical fibers for communication use are mostly fabricated from *

1/1

- ☐ a) Plastic
- ☒ b) Silica or multicomponent glass
- ☐ c) Ceramics
- ☐ d) Copper



✓ The _____ ray passes through the axis of the fiber core *

1/1

- ☐ a) Reflected
- ☐ b) Refracted
- ☒ c) Meridional
- ☐ d) Shew



✗ P2O5 is used as a *

0/1

- ☐ a) Dopant
- ☐ b) Starting material
- ☒ c) Cladding glass
- ☐ d) Core glass



Correct answer

- ☒ a) Dopant

✓ The fibers mostly not used nowadays for optical fiber communication system are

*1/1

- ☒ a) Single mode fibers
- ☐ b) Multimode step fibers
- ☐ c) Coaxial cables
- ☐ d) Multimode graded index fibers



✓ The phenomenon leading to avalanche breakdown in reverse-biased diodes is known as _____. *1/1

- ☐ a) Auger recombination
- ☐ b) Mode hopping
- ☒ c) Impact ionization
- ☐ d) Extract ionization



✓ _____ converts the received optical signal into an electrical signal. * 1/1

- ☒ a) Detector
- ☐ b) Attenuator
- ☐ c) Laser
- ☐ d) LED



✓ Fiber mostly suited in single-wavelength transmission in O-band is? * 1/1

- ☐ a) Low-water-peak non dispersion-shifted fibers
- ☒ b) Standard single mode fibers
- ☐ c) Low minimized fibers
- ☐ d) Non-zero-dispersion-shifted fibers



✓ An optical fiber has core-index of 1.480 and a cladding index of 1.478. *1/1
What should be the core size for single mode operation at 1310nm?

- ☐ a) 7.31 μ m
- ☐ b) 8.71 μ m
- ☐ c) 5.26 μ m
- ☒ d) 6.50 μ m



SUBJECT NAME *

- ☒ Optoelectronics & Optical Communication

✓ The optical bandwidth is _____ the electrical bandwidth. * 1/1

- ☐ a) Smaller
- ☒ b) Greater
- ☐ c) Same as
- ☐ d) Zero with respect to



✓ In photo detectors, energy of incident photons must be _____ *1/1
band gap energy.

- ☐ a) Lesser than
- ☒ b) Greater than
- ☐ c) Same as
- ☐ d) Negligible



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